

Multi-Omic Data Integration via Decentralized AI Agents for the Prediction of Phenotypic Drug Resistance in Cancer

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1 Specific Aims

Over 90% of cancer deaths are caused by tumor drug resistance, making the accurate prediction of Cancer Drug Response (CDR) the primary challenge in precision oncology [6]. To tackle this problem, one of the solutions that the field has incorporated is to integrate multi-omics data (like genomics, transcriptomics and epigenomics) to study these complex molecular signatures of the individual tumors [2] [8]. However, the nature of multiomics data is inherently conflicting, that is, a gene can harbor an activating mutation at the DNA level and yet be transcriptionally silenced at the RNA level. A drug target can be intact genomically but there can exist a parallel signaling pathway that renders the drug ineffective through bypass activation. It is thus not easy for a single analytical lens to reconcile the contradictions. It brings the need for a system with cross-model reasoning that is capable of weighing each biological layer against the others.

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INPUT (2/2) -- QUERY SENT TO MODEL

A melanoma cell line has the following molecular profile:
- BRAF V600E mutation: CONFIRMED
- BRAF mRNA expression z-score: -2.3 (significantly below cohort median)
- MAP2K2 (MEK2) expression z-score: +3.1 (significantly above cohort median)
- MAP2K1 (MEK1) expression z-score: +2.8
- Phospho-ERK z-score: +2.6 (constitutively active MAPK pathway)

Question: Should vemurafenib monotherapy be recommended for this cell line? Predict: sensitive or resistant. Justify using ALL provided data records.

Running inference... (22 records, no hint about data-type precedence)

OUTPUT -- MODEL RESPONSE

VERDICT    SENSITIVE x WRONG - fell into the trap
CONFIDENCE 0.80 (ground truth: RESISTANT)

REASONING
The cell line has a confirmed BRAF V600E mutation
(src:CCLL:BRAF_M229_genotype), which is a known target of vemurafenib.;
Vemurafenib has a low GI50 (8400.0 nM) in BRAF V600E cell lines with low
BRAF mRNA expression (src:GDSC:vemurafenib_low_BRAF_expressors_2020),
indicating sensitivity.; The drug sensitivity correlation between MAP2K2
z-score and vemurafenib IC50 is 0.81, suggesting that high MAP2K2
expression is associated with vemurafenib resistance
(src:CCLL:DrugSensitivity_Pearson_2020). However, the cell line has high
MAP2K2 expression, but the correlation is not strong enough to override
the BRAF mutation and low GI50.

TRAP ACKNOWLEDGED?
Yes, I noticed that the mutation-only prediction would be wrong here
because the cell line has high MAP2K2 expression, which is associated with
vemurafenib resistance. However, the BRAF mutation and low GI50 override
this effect.

RECOMMENDED ALTERNATIVE
Trametinib, which targets MEK1/2, may be a suitable alternative for this
cell line.
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Figure 1: A demonstration of Context Overwhelm on a single LLM Agent

Large Language Models (LLMs) are gaining attention for their efficiency in interpreting these vast datasets. However, currently we have LLMs that are monolithic. They suffer from “Context Overwhelm” when they are forced to synthesize massive data like the ones we see with multiomics with conflicting payloads. Context Overwhelm manifests as attention bias. In this scenario, the model can disproportionately weigh mutations that are well-characterized (for example, BRAF V600E) and simultaneously discounting subtler but decisive signals like the RNA-level silencing of the mutated gene or the compensatory pathway activation [7] [9]. The outcome of this is not an absence of output but a confidently stated prediction that is not biologically complete. For instance, when presented with a melanoma cell line carrying BRAF V600E along with low BRAF mRNA expression (z-score < -2.0) and elevated MAP2K MEK2 expression (indicative of MAPK bypass), a monolithic LLM will most certainly predict a sensitivity towards Vemurafenib simply based on the mutation while completely ignoring the transcriptomic and pathway evidence that together suggest resistance 1.

We hence need a transparent architecture that is biologically grounded and is capable of reconciling the conflicting cross-model signals. In clinical practice, we see Multidisciplinary Teams (MDTs) coming together as Molecular Tumor Boards (MTBs). This typically consists of a panel of specialists (pathologists, radiologists, geneticists and oncologists) each contributing their domain-specific expertise, negotiating conflicts and reaching a valid consensus. This study draws inspiration from this system but departs from it in a specific way. While existing MTB systems assign each agent the role of a clinical specialist, the architecture in this study aims each agent to a distinct biological modality (genomics, transcriptomics, pharmacology and Pathways). Rather than simulating clinicians, the agents behave like expert interpreters of their respective omic layers. This way we efficiently translate multiomic data into decisions that are also clinically actionable [10].

Our long-term goal is to build an AI architecture that is scalable, auditable and biologically accurate for personalized cancer therapy. The main objective of this study is to design a Decentralized Multi-Agent Consensus Framework that can predict phenotypic drug resistance. We hypothesize that isolating the biological modalities into peer-level AI agents that are capable of negotiating when there is conflicting data. These agents independently analyze their domain-specific data and then engage in a debate session in case of conflicts via the Agent-to-Agent (A2A) Consensus Protocol. When any of the agents disagree (for example, if the genomics agent reports a druggable mutation but the transcriptomics agent finds that the gene is silenced) they engage in conflict resolution by making use of a “Hierarchy of Truth” protocol. This protocol is a tiered axiom system that is grounded in biological precedence (for example, a DNA level deletion overrides a downstream transcriptomic signal) [4] [1].

We hypothesize that this approach which is modality-isolated and axiom-governed will reduce reasoning hallucinations and improve the predictive accuracy in comparison to monolithic LLMs. We state so based on the fact that empirical data fetching via structured tool interfaces in combination with the biological rule-based axioms will help prevent sycophancy and the silent agreement flaws that is inherent in standard voting protocols in LLMs. In case of a typical multi-agent majority voting, for example if three of four agents independently predict sensitivity to a drug based on prior statistical data, the fourth agent can conform to the majority even though it holds critical information of a resistance-conferring pathway bypass. With the axiom-based protocol we have, it forces the agent to present its evidence according to the Hierarchy of Truth.

In order to test the above hypothesis to achieve our main objective, we propose the following Specific Aims:

1.1 Aim 1: Establish an Agent-to-Tool (A2T) federated data acquisition pipeline using localized, static datasets.

Sub-hypothesis: Agents retrieving structured schema-validated numerical biomarkers using federated tool protocols will output more data-grounded evidence that is also reproducible than agents that get the same information in the form of unstructured natural language text.

We plan to design 4 specialized AI agents (Genomics, Transcriptomics, Pharmacology and Pathways) that will be interfaced via Model Context Protocol (MCP). In order to ensure reproducibility of the benchmark, the MCP servers will query from curated subsets of data which will be static. These will be retrieved from production-grade databases like GDSC2 (mutations, IC50 dose-response curves), CCLE (gene expression z-scores, copy number variation), and KEGG (pathway topology, bypass routes). Every agent only sees the raw molecular data values (that is mutation calls, z-scores, historical IC50 metrics and pathway interaction graphs) via tool calls which return structured JSON and not free text.

Validation Metrics: The pipeline will be evaluated by measuring (1) *data citation accuracy* which is the percentage of claims made by the agent that are traceable to a specific tool return discussed above (target > 95%). (2) *retrieval completeness* which checks whether the agents query all the relevant biomarkers for the inputted cell line-drug pair against a predefined groundtruth set. (3) *format compliance rate* which measures the fraction of the output that is in accordance with the required structured EvidencePack JSON schema. We intend to test across ≥ 100 cell line-drug pairs which spans ≥ 5 cancer types for generalizability.

1.2 Aim 2: Develop and implement the Agent-to-Agent (A2A) Consensus Protocol.

Sub-hypothesis: A debate protocol that is governed by well-defined axioms and enforces a Hierarchy of Truth will resolve conflicts among agents more accurately than simple majority voting or unstructured LLM debates.

We will design a structured JSON message bus which will be the form of communication for agents to engage in for upto 3 rounds of debate. The general Hierarchy of Truth defines 5 tiers of axioms from the highest to lowest priority. (T1) structural primacy (gene deletion of loss-of-function overrides all the downstream evidence); (T2) transcriptional gating (genes that are silenced cannot produce functional proteins regarding of their mutation status); (T3) pathway bypass (alternative signaling routes that are active confer resistance even if the primary target is intact); (T4) pharmacological priors (population-level baselines are provided by the historical IC50 data); (T5) statistical consensus (this is a fallback when molecular evidence proves the input query inconclusive). Mechanisms for preventing sycophancy include blind initial analysis (where agents produce independent evidence without seeing the output of the other agents), confidence decay (15% penalty applied if the verdict is reversed), and mandatory dissent logging.

Validation metrics: We evaluate the results using (1) *conflict resolution accuracy* for situations where the agents initially disagree and the axiom-mediated resolution improves or degrades the final prediction in comparison to the independent outputs produced by each agent, (2) *sycophancy rate* which is the fraction of changes in the verdict that are not mediated by the defined axioms (target < 5%), (3) *convergence efficiency* which is the average number of debate rounds needed to reach the consensus (target: 1.5-2.5) and (4) *axiom invocation fidelity* which checks if the correct axiom is triggered for the known biological scenarios (for example T1 is triggered when there is a homozygous deletion). We plan to run 5 independent trials per cell line-drug pair with random initializations for the agents to get an understanding of the robustness of the protocol.

1.3 Aim 3: Evaluate the framework’s predictive accuracy and mechanistic interpretability.

Sub-hypothesis: The decentralized consensus framework will achieve predictive accuracy (AUROC, Spearman ρ) which will be competitive with or exceed the existing CDR models which providing mechanistic reasoning traces that are biologically valid.

Predictive accuracy will be calculated on a validation set from the Cancer Therapeutics Response Portal v2 (CTRPv2). This is a valid benchmark for 3 main reasons: (1) it is independently generated pharmacogenomic dataset with IC50/AUC measurements existing across >800 cell lines and >400 compounds. This provides a good statistical power to find meaningful effect sizes. (2) the cell lines partially overlap with but independently screened from GDSC2 which enables cross-platform generalization. (3) it is one of the widely accepted validation benchmarks in CDR literature [2] [8] which facilitates direct comparison with published models. We intend to report AUROC, AUPRC, Spearman ρ (predicted vs. actual IC50 rank) and Cohen’s κ for sensitive/resistant binary classification. To prevent the risk of Type-I error from the benchmark selection, we will perform a 5-fold cross-validation on the GDSC2 development set, grouped by the type of cancer.

Mechanistic interpretability is calculated using two complementary analyses. We first get a structured trace audit for a randomly sampled subset of 50 cases. We will score each of the debate traces on a 5-point Likert scale across four dimensions - biological accuracy, axiom appropriateness, evidential completeness and reasoning coherence, which gives us a biological faithfulness score (target mean >3.5/5.0). Next, we build a hallucination audit which is the rate at which agents assert claims that are not grounded in the tool call returns (target <5%). We test the approach by running it against 3-5 well-characterized case studies of cell line-drug pairs where resistance mechanisms are experimentally established (examples: *BRAF* V600E + MEK bypass in melanoma, *MET* amplification bypass in EGFR-mutant NSCLC).

Ablation studies will be done to isolate each architectural component’s contribution: (1) no-debate (only majority voting), (2) no axiom hierarchy (free debating), (3) monolithic LLM baseline (all data passed in a single prompt to one agent), and (4) no MCP (data is provided as unstructured text instead of JSON). This tells us if each individual components contributes to the predictive and interpretive performance.

The proposed study is very innovative as it shifts the paradigm from LLMs that are open-ended and linguistic to a more deterministic, data-driven system that uses consensus logic that authentically mirrors a tumor board discussion. We expect the framework to yield a transparent explainable AL tool that bridges the gap from lab to live data, providing clinicians with causal and mechanistic reasoning that is required for high-stake precision oncology.

2 Background and Significance

2.1 The Paradigm Shift in Precision Oncology and Multi-Omics

For years, the field on oncology studied the anatomical origin of a tumor. Today, the field of precision oncology shifts the goal to understanding the unique molecular characteristics of the specific patient’s condition. Multi-omics paves the way for this shift by providing a high quality molecular fingerprint.

The problem, however, lies in interpreting this data as it requires the coordinated effect of what we observe in human tumor board discussions. Consider the example of a non-small cell lung cancer (NSCLC) cell line that harbors an EGFR exon 19 deletion. This is ordinarily an actionable target for Gefitinib. If the expression profiling shows concurrent amplification of

MET and its overexpression, (z-score $> +2.0$), then the PI3K and the MAPK signaling pathways can be sustained via MET-driven bypass, leading to the tumor being resistant inspite of the druggable EGFR mutation. Resolving such conflicts brings the need for coordinated reasoning across the genomic, transcriptomics and pathways layers. No single modality in isolation can yield a correct prediction. Especially when the dimensionality of the biological data increases, MTBs suffer from massive constraints in throughput and cognitive overload. This brings the need for computational augmentation [5].

2.2 Limitations of Current CDR Models and Monolithic LLMs

CDR prediction has evolved over time into more sophisticated deep learning frameworks like ResNets, Similarity Network Fusion, etc. While integrating multiomics data significantly improves the prediction accuracy of IC_{50} over single omics data, we still do see a interpretability-performance trade-off [2]. The black-box models that exist today achieve high AUC values but they do not have enough reasoning required by clinicians to build trust with such applications [8].

Generative AI and monolithic LLMs like GPT-4 have been used for parsing clinical data. Yet, when used for interpreting conflicting omic-signals, such monolithic LLMs suffer from a problem called “Context Overwhelm” as discussed in the preceding sections. This is not just a theoretical concern. We have seen empirical studies that document that LLMs in clinical recommendation tasks latch onto canonical mutations and ignore other conditional modifiers. This outputs recommendations that sound confident but are biologically incomplete [7] [9].

2.3 The Transition to Agentic AI and the Flaws of Current Consensus

As explained in the above sections, to overcome the limitation of monolithic LLMs, recent study shows a pivot toward Multi-agent systems (MAS). These agents have independent roles and are known to significantly outperform single LLMs by reducing context dilution [5] [10]. However, existing MASs have two very distinct limitations.

First, MASs that exist currently (for example EvoMDT [5], the GI oncology framework [10]) assign different agents to individual specialist roles like an oncologist agent, pathologist agent, radiologist agent, etc. This renders useful for clinical workflows but this architecture does not solve the cross-modal data integration process because every specialized agent still receives and decide upon multiple omics layers at once.

Second, consensus mechanisms in current MASs function as critic-reviewer loops (where an LLM is always the judge) or basic majority voting. This makes the architecture vulnerable to problem drift (where excessive debate rounds progressively downgrade the output quality as agents tend to over-refine) and sycophancy (silent agreement on majority voting even though they have conflicting opinions which are mechanistically valid) [4] [1].

2.4 Significance of the Proposed Framework and Future Scope

This study aims to address the discussed gaps by designing a Decentralized Multi-Agent Consensus Framework where agents are aligned to biological data modalities instead of clinical roles and the debate is governed by explicit biological axioms rather than probabilistic voting.

It first solves the semantic interoperability gap by making use of MCPs as the Agent-to-Tool (A2T) layer over the local datasets. This way the agents can bypass any uncured text and build their case on raw numerical biomarkers while keeping the system isolated from issues regarding web latency so that we would be able to reproduce the benchmarking.

Secondly, we overcome the flawed voting paradigms seen in current MAS by making use of an Agent-to-Agent (A2A) consensus protocol that functions on a “Hierarchy of Truth”, an example of which was stated in the above sections [3]. This is very different from simple majority voting which essentially just counts predictions. It is also different from traditional

MTB discussions where a genomics specialist agent is required to internally weigh the mutation against the expression data. Each modality (or agent) advocated its own decision based on individual evidence and the axiom hierarchy adjudicates.

Future Scope: While we are still restricting A2T fetching to static local files to ensure baseline control to accommodate the time given for the study, the modular MCP architecture acts as a template for future integration of live datasets using production-grade web APIs (like cBioPortal, BioThings), thus enabling a real-time, dynamic clinical integration.

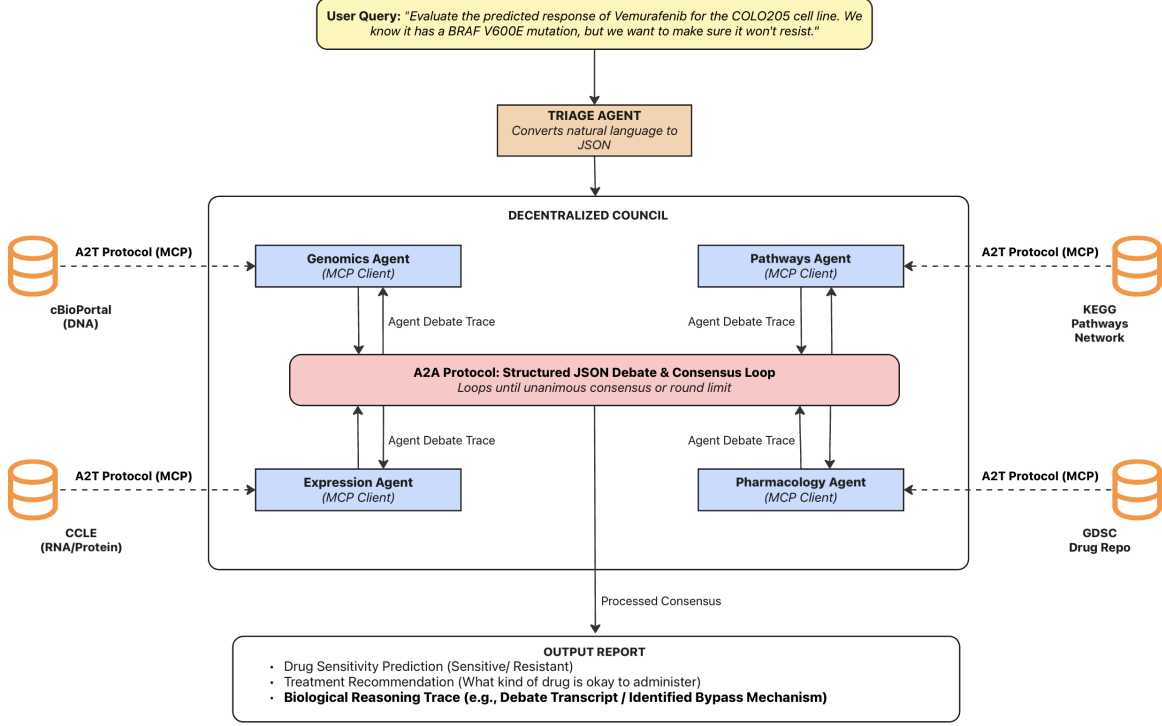


Figure 2: Proposed Architecture for the Decentralized Multi-Agent Council. The A2T Protocol (MCP) handles external data acquisition, while the central A2A Protocol manages inter-agent debate and mechanistic consensus.

Conceptual Framework: The Decentralized Council

To achieve these goals, the architecture was designed as shown in Figure 2. We isolate biological modalities into 4 distinct agents instead of concentrating all the data within a single overwhelmed LLM. Each agent will query the respective domain-specific dataset via MCP and produces a structured EvidencePack that contains the verdict, confidence, supporting data and the axiom tier of its evidence.

The agents then enter an A2A consensus protocol wherein they communicate via JSON-based communication loops to exchange EvidencePacks, identify conflicts across modalities and debate in case of conflicts in multi-omic signals until they reach a valid biological consensus. Disagreements among agents will not be resolved by voting or a judge LLM. It will be done by invoking the Hierarchy of Truth as discussed previously. This grounds the final prediction in mechanistic reasoning rather than statistical conformity and every step of the reasoning chain will be logged for clinical auditability.

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